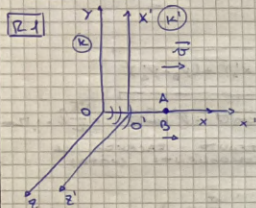


④

TORNADO: 2, 9
ESERCIZI: 57 - 72

TORNADO FISICA III

ESERCIZIO IN DUE



$$\begin{aligned}x_A &= L & x_B(t=0) &= L \\y_A &= 0 & y_B(t=0) &= 0 \\z_A &= 0 & z_B(t=0) &= 0\end{aligned}$$

POSSIAMO SCRIVERE DAL TESTO $x_B(t) = L + vt$

$$a) \quad ct_A = L \Rightarrow t_A = \frac{L}{c}$$

$$ct_B = L + vt_B \Rightarrow t_B = \frac{L}{c-v} = \frac{L}{c} \frac{1}{1-\beta}$$

POSIZIONE DI A QUANDO ARRIVA IL SEGNALE $d_A = L$

$$d_B = L + vt_B = \frac{L}{1-\beta}$$

b) USIAMO LE TRASFORMAZIONI DI LORENTZ

$$d'_A = \gamma(d_A - vt_A) = L\sqrt{1-\beta^2}$$

$$d'_B = \gamma(d_B - vt_B) = \gamma L$$

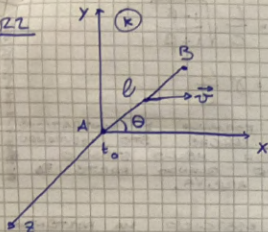
$$t'_A = \gamma(t_A - \frac{v}{c^2}x_A) = \frac{L}{c} \sqrt{1-\beta^2}$$

$$t'_B = \gamma(t_B - \frac{v}{c^2}x_B) = \gamma \frac{L}{c}$$

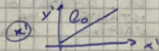
c) NON VALE LA FORMULA DI CONTRAZIONE DELLE LUNGHEZZE PERCHÉ STO MISURANDO LE LUNGHEZZE NON NELLO STESSO ISTANTE MA UNO QUANDO PARTE E L'ALTRO QUANDO ARRIVA.

d) PROVA STESSO ESERCIZIO MA PARTENDO DAL TROVARE QUELLI IN K' E POI RICAVARE QUELLI IN K

22



NOTA
SO HO LA SBARRA
FERMA IN K' ED
LO È LUNGHEZZA
PROPRIA E VOGLIO
LA LUNGHEZZA
IN UN SIZ IN ROTO
SCOMPONGO LE
COMPONENTE E USO
LORENTZ



$$l = 1,0 \text{ m}, \quad \theta = 30^\circ, \quad v = \frac{c}{2}$$

$$\text{CALCOLO } \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{2}{\sqrt{3}}$$

a t_0 è: ~~la posizione degli estremi nello stesso istante~~

$$t_0 \begin{cases} x_A = 0 \\ y_A = 0 \\ z_A = 0 \\ t_A = t_0 \end{cases} \quad \begin{cases} x_B = l \cos \theta \\ y_B = l \sin \theta \\ z_B = 0 \\ t_B = t_0 \end{cases}$$

PRIMO LA POSIZIONE DEGLI ESTREMI NELLO
STESSO ISTANTE

USIAMO LORENTZ

$$\begin{cases} x'_A = \gamma(x_A - vt_A) = -\gamma vt_0 \\ y'_A = y_A \\ z'_A = z_A \end{cases}$$

t'_A NON MI SERVE...

$$\begin{cases} x'_B = \gamma(x_B - vt_B) = \gamma(l \cos \theta - vt_0) \\ y'_B = y_B = l \sin \theta \\ z'_B = z_B \end{cases}$$

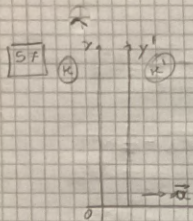
$$\text{CALCOLIAMO } l_0 = \sqrt{(x'_B - x'_A)^2 + (y'_B - y'_A)^2} \\ = \sqrt{l^2 \gamma^2 \cos^2 \theta + l^2 \sin^2 \theta} = \frac{\sqrt{3}}{2} l = 1,12 \text{ m}$$

STESSO RISULTATO FACENDO LA
SCOMPOSIZIONE IN COMPONENTI (NOTA)

$$\text{VALE } t'_A = \gamma(t_A - \frac{v}{c^2} x_A)$$

SO $t_A = t_B$ MA $x_A \neq x_B$ QUINDI $t'_A \neq t'_B$

MA NON IMPORTA PERCHÉ TANTO
 $x'_A \neq x'_B$ SONO COSTANTI IN K'
PERCHÉ LA SBARRA È FERMA



K' MISURA L. SEGNALC IN M, N, P

$$\boxed{\kappa} \quad t_n = \frac{d}{c} \quad t_o = \frac{d}{c} \quad t_p = \frac{d}{2c}$$

$$\begin{aligned} \text{a) } t'_n &= \gamma(t_n - u \frac{d}{c^2}) = \gamma(\frac{d}{c} - u \frac{d}{c^2}) \\ &= \gamma \frac{d}{c} (1 - \frac{u}{c}) = \gamma \frac{d}{c} (1 - \beta) \end{aligned}$$

$$t'_o = \gamma(t_o + u \frac{d}{c^2}) = \gamma(\frac{d}{c} + u \frac{d}{c^2}) = \gamma \frac{d}{c} (1 + \beta)$$

$$t'_p = \gamma(t_p - 0) = \gamma \frac{d}{2c}$$

$$\text{b) } t'_n = t'_p \Rightarrow \gamma \frac{d}{c} (1 - \beta) = \gamma \frac{d}{2c}$$

$$\Rightarrow 1 - \beta = \frac{1}{2} \Rightarrow \beta = \frac{1}{2} \Rightarrow u = \frac{c}{2}$$

$$\text{c) } t'_o = t'_p \Rightarrow \gamma \frac{d}{c} (1 + \beta) = \gamma \frac{d}{2c} \Rightarrow u = -\frac{c}{2}$$

$$\text{d) } \Delta t = \frac{4d}{2c} = t_n - t_o \Rightarrow \gamma(\frac{d}{c} - u \frac{d}{c^2}) - \gamma \frac{d}{c} (1 + \beta) = \frac{4d}{2c}$$

$$t_o = 0$$

$$M = (d, 0, 0)$$

$$N = (-d, 0, 0)$$

$$P = (0, \frac{d}{2}, 0)$$

$$d > 0$$

$$\gamma \frac{d}{c} [1 - \beta - 1 + \beta] = \frac{4d}{2c}$$

$$-2\beta\gamma = \frac{4}{3} \Rightarrow \beta \frac{1}{\sqrt{1-\beta^2}} = -\frac{2}{3}$$

$$\frac{u^2}{c^2(1-\frac{u^2}{c^2})} = -\frac{2}{3} \quad \frac{u^2}{(c^2-u^2)} = \frac{4}{9}$$

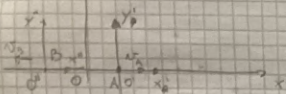
$$\frac{u^2}{c^2-u^2} = \frac{4}{9} \quad u^2 = \frac{4}{9}(c^2-u^2)$$

$$u^2(1 + \frac{4}{9}) = \frac{4c^2}{9}$$

$$u^2 = \frac{4c^2}{9} \frac{9}{9+4} = \frac{4c^2}{13}$$

$$u = \frac{\pm 2c}{\sqrt{13}} = \pm \frac{2c}{\sqrt{13}}$$

50 m



$$\vec{v}_A = \frac{3}{5} c \hat{u}_x$$

$$\vec{v}_B = -\frac{4}{5} c \hat{u}_x$$

a)
$$x'_A = \gamma (x_A - v_A t) = \gamma (x_A - \frac{3}{5} c t)$$

$$x'_A = x_A$$

$$t'_A = t_A$$

$$t'_A = \gamma (t_A - v_A \frac{x_A}{c}) = \gamma (t_A - \frac{3}{5} \frac{x_A}{c})$$

$$\beta_A = \frac{v_A}{c} = \frac{3}{5}$$

$$\beta_B = -\frac{4}{5}$$

$$\gamma_A = \frac{1}{\sqrt{1-\beta_A^2}} = \frac{1}{\sqrt{1-\frac{9}{25}}} = \frac{5}{4}$$

$$\gamma_B = \frac{5}{3}$$

$$x'_A = \frac{5}{4} (x_A - \frac{3}{5} c t) \quad t'_A = \frac{5}{4} (t_A - \frac{3}{5} \frac{x_A}{c})$$

b)
$$x'_B = \gamma (x_B - v_B t) = \frac{5}{3} (x_B + \frac{4}{5} c t)$$

$$x'_B = x_B$$

$$t'_B = t_B$$

$$t'_B = \gamma (t_B - v_B \frac{x_B}{c}) = \frac{5}{3} (t_B + \frac{4}{5} \frac{x_B}{c})$$

$$x'_B = \gamma (x'_B + \beta c t'') = \frac{5}{3} (x'' - \frac{4}{5} c t'')$$

$$x'_B = x'' \quad t'_B = t''$$

$$c t'_B = \gamma (c t'' + \beta x'') = \frac{5}{3} (c t'' + \frac{4}{5} x'')$$

c) so
$$x = \frac{5}{3} (x'' - \frac{4}{5} c t'')$$

$$y = y''$$

$$z = z''$$

$$t = \frac{5}{3} (t'' + \frac{4}{5 c} x'')$$

INVERSE
DA B A (K)

$$x' = \frac{5}{4} (x - \frac{3}{5} c t)$$

$$x' = x$$

$$t' = t$$

$$t' = \frac{5}{4} (t - \frac{3}{5} \frac{x}{c})$$

DA (K) A (L)

TROVO COMBINANDO

$$x' = \frac{5}{4} (\frac{5}{3} (x'' - \frac{4}{5} c t'') - \frac{3}{5} c (\frac{5}{3} (t'' + \frac{4}{5 c} x''))) =$$

$$x' = x''$$

$$t' = t''$$

$$t' = \frac{5}{4} (\frac{5}{3} (t'' + \frac{4}{5 c} x'') - \frac{3}{5 c} (\frac{5}{3} (x'' - \frac{4}{5} c t''))) =$$

$$x' = \frac{5}{4} (\frac{5}{3} x'' - \frac{4}{3} c t'' - c t'' + \frac{4}{3} x'') =$$

$$= \frac{5}{4} (\frac{15+12}{15} x'' - \frac{7}{3} c t'') = \frac{37}{12} (x'' - \frac{35}{37} c t'')$$

$$t' = \frac{5}{4} \left(\frac{5}{3} t'' - \frac{4}{3c} x'' - \frac{1}{c} x'' + \frac{4}{5} t'' \right)$$

$$= \frac{5}{4} \left(\frac{25+12}{15} t'' - \frac{7}{3c} x'' \right) = \frac{37}{12} \left(t'' - \frac{35}{37} \frac{1}{c} x'' \right)$$

$$d) t' = \frac{37}{12} \left(t'' - \frac{35}{37} \frac{1}{c} x'' \right) = \gamma \left(t'' - \frac{v}{c^2} x'' \right)$$

$$\Rightarrow \frac{35}{37} \frac{1}{c} x'' = \frac{v}{c^2} x'' \Rightarrow v = \frac{35}{37} c \hat{u}_x$$

$$\gamma = \frac{37}{12} = \frac{1}{\sqrt{1-\beta^2}} \Rightarrow \left(\frac{37}{12} \right)^2 = \frac{1}{1-\beta^2}$$

$$1-\beta^2 = \left(\frac{12}{37} \right)^2$$

$$\beta^2 = 1 - \left(\frac{12}{37} \right)^2 \Rightarrow \beta = \sqrt{1 - \left(\frac{12}{37} \right)^2}$$

$$v = c \sqrt{1 - \left(\frac{12}{37} \right)^2} = c \sqrt{\frac{1269 - 144}{37^2}}$$

$$\boxed{v = c \frac{25}{37}}$$

REGGE COMPOSIZIONE DELLE VELOCITÀ

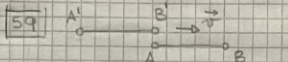
$$u' = \frac{u'' - v}{1 - \frac{v u''}{c^2}} = c^2 \frac{u'' - v}{c^2 - v u''}$$

$$u' (c^2 - v u''_x) = c^2 (u''_x - v)$$

$$u' (c^2 - v u''_x) = c^2 (u''_x - v)$$

$$u' = c^2 \frac{u''_x - v}{c^2 - v u''_x} = \frac{u''_x - v}{1 - \frac{v u''_x}{c^2}} = + \frac{25}{37} c$$

$$\boxed{u''_x = v \quad u''_x = v}$$



OROLOGI SINCROINIZZATI A-B A'-B'

$$a) \begin{cases} x_A = 0 & x_B = l_0 \\ x_A = v t_A = l & x_B = v t_B \end{cases}$$

USPETTO $\odot$

$$x_B = x_A$$

$$x_B = l_0 = v t_B \rightarrow t_B = \frac{l_0}{v}$$

$$x_A = x_A \rightarrow v t_A - l = 0 \rightarrow t_A = \frac{l}{v}$$

CONTRAZIONE LUNGHEZZE

$$t_A = \frac{l_0}{v}$$

USO LORENTZ

$$t'_A = \gamma \left(t_A - \frac{v}{c^2} x_A \right) = t'_A = \gamma t_A = \frac{l_0}{v}$$

$$t'_B = \gamma (c t_B - \beta x_B) = \gamma \left(c \frac{l_0}{v} - \beta l_0 \right) = \gamma \left(c \frac{l_0}{v} - \beta \frac{l_0}{v} \right)$$

$$= \gamma \frac{l_0}{v} (c - \beta v) = \gamma \frac{l_0}{v} c (1 - \beta \frac{v}{c}) = \gamma \frac{l_0}{v} c (1 - \beta^2)$$

$$= \gamma \frac{l_0}{v} c \frac{1}{\gamma^2} = \frac{l_0}{v} c \Rightarrow t'_B = \frac{l_0}{v}$$

here and Antarctic La
(FRANCE)

60 K rispetto K $v_1 = \frac{c}{\gamma}$
 K' rispetto K' $v_2 = \frac{c}{\gamma}$

$t=0$ TUTTI INSIEME
 $K \rightarrow K'$
 $K \rightarrow K''$
 $\gamma_1 = \frac{1}{\sqrt{1-\frac{1}{\gamma^2}}} = \gamma$
 $\gamma_2 = \frac{1}{\sqrt{1-\frac{1}{\gamma^2}}} = \gamma$

$\beta_1 = \frac{v_1}{c} = \frac{1}{\gamma}$ $\beta_2 = \frac{1}{\gamma}$

VALE FASQIO DE LUCE

$x = ct$ $x' = ct'$ $x'' = ct''$

VALE LORENTZ

$ct' = \gamma(ct - \frac{v_1}{c}x)$ $ct'' = \gamma(ct' - \frac{v_2}{c}x')$
 $y' = y$ $y'' = y'$
 $z' = z$ $z'' = z'$
 $x' = \gamma_1(x - v_1 t)$ $x'' = \gamma_2(x' - v_2 t')$

$x' = \gamma_1(x - v_1 t)$
 $x'' = \gamma_2(\gamma_1(x - v_1 t) - v_2 \gamma_1 t)$
 $= \gamma_2 \gamma_1 (x - v_1 t - v_2 t)$
 $= \frac{\gamma}{\gamma} (x - (v_1 + v_2) t)$
 $= \frac{\gamma}{\gamma} (\frac{c}{\gamma} x - \frac{c}{\gamma} ct) = \frac{c}{\gamma} (x - \frac{c}{\gamma} ct)$

$y'' = y$ $z'' = z$

$$\begin{aligned}
 ct'' &= \gamma_{11} \left(\gamma_1 \left(ct - \frac{v_1}{c} x \right) - \frac{v_2}{c} \gamma_1 \left(x - v_1 t \right) \right) \\
 &= \gamma_{11} \gamma_1 \left(ct - \frac{v_1}{c} x - \frac{v_2}{c} x + \frac{v_1 v_2}{c} t \right) \\
 t'' &= \frac{\gamma}{\sqrt{\beta}} \left(x \left(\frac{v_1}{c} + \frac{v_2}{c} \right) + t \left(1 + \frac{v_1 v_2}{c^2} \right) \right) \\
 &= \frac{\gamma}{\sqrt{\beta}} \left(t \frac{7}{5} - x \frac{5}{5c} \right) = \frac{7}{2\sqrt{\beta}} \left(t - \frac{5}{7} \frac{x}{c} \right)
 \end{aligned}$$

SO LO TRASFORMAZIONI DI LORENTZ
NELLA FORMA

$$t' = \gamma \left(t - \frac{v_0 x}{c^2} \right)$$

quindi da $t'' \rightarrow v = \frac{5}{7} c$

01 $\tau_0 = 10^{-8} s$ in K'
in K qual è il Δx se
 $L = \frac{v}{c} = 2 \cdot 10^{-8} s$

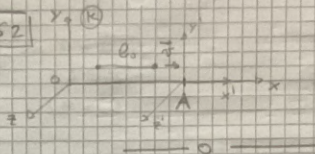
SILTAZIONE DEI TEMPI $\tau = \gamma \tau_0$

$$\Rightarrow \gamma = \frac{\tau}{\tau_0} = 2$$

$$\gamma = \frac{c}{\sqrt{c^2 - v^2}} \rightarrow \sqrt{c^2 - v^2} = \frac{c}{2} \rightarrow v = \sqrt{c^2 - \frac{c^2}{4}}$$

$$\rightarrow v = \frac{\sqrt{3}c}{2} \rightarrow d = v \tau = 5.19 m$$

62 $\gamma_0 (K)$



$$\Delta t = 20 ns$$

$$\Delta t' = 25 ns$$

Δt TEMPO PROPRIO

DETAZIONE DEI TEMPI

$$\Delta t' = \gamma \Delta t$$

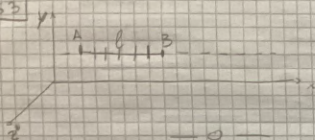
$$\Rightarrow \gamma = \frac{\Delta t'}{\Delta t} = \frac{5}{4}$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \rightarrow v = c \sqrt{1 - \frac{1}{\gamma^2}} = \frac{3c}{5}$$

$$L_0 = v \Delta t = 4.5 m$$

TEMPO PER ATTRAVERSARLA TUTTA

63



$$l = 1 \text{ mm}$$

$$l = 45$$

$$l' = 95$$

(B2° note)

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} \Rightarrow \sqrt{1 - \beta^2} = \frac{1}{\gamma}$$

$$\Rightarrow \beta = \sqrt{1 - \frac{1}{\gamma^2}} \quad \text{and} \quad x = c \sqrt{1 - \frac{1}{\gamma^2}} = \frac{\sqrt{5}}{3} c$$

$$x_A = 0 \quad x_B = 45$$

$$x'_A = 0 \quad x'_B = 0$$

$$t_A = 0 \quad t_B = 0$$

$$t'_A = 0 \quad t'_B = 0$$

TRADE CONTRAZIONE LUNGHEZZE

USO LORENTZ

$$x'_A = 0 \quad x'_B = \gamma x_B$$

$$x'_A + x'_B = x'_B = \gamma x_B \Rightarrow \gamma = \frac{x'_B}{x_B}$$

$$\gamma = \frac{5}{3}$$

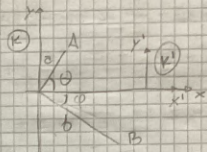
$$x'_B = 95$$

$$\gamma = \frac{95}{45} \Rightarrow \gamma = \frac{19}{9}$$

CONTRAZIONE LUNGHEZZE

$$L = \gamma L_0 = \frac{1}{\gamma} L_0 = \frac{1}{19/9} 45 = 21 \text{ mm}$$

64



$$\varphi = \frac{\pi}{2} - \theta$$

$$a \neq b$$

TRADE K' to a-b

VOGLIO $|a| = |b|$ in K'

$$|a_x| = a \cos \theta$$

$$a_y = a \sin \theta$$

$$|b_x| = b \cos \varphi = b \sin \theta \quad b_y = b \cos \theta$$

SO CHO COMPONENTE TRASVERSE AD X SI CONSERVANO

$$a'_y = a \sin \theta = b_y \quad b'_y = b \cos \theta = b_y$$

V VELOCITA DI K'

CONTRAZIONE LUNGHEZZE

$$a'_x = \frac{a_x}{\gamma} \quad b'_x = \frac{b_x}{\gamma}$$

E POI IMPONENDO $|a'| = |b'|$

$$\sqrt{\frac{a_x^2}{\gamma^2} + a_y^2} = \sqrt{\frac{b_x^2}{\gamma^2} + b_y^2}$$

$$a^2 \cos^2 \theta + a^2 \sin^2 \theta = b^2 \sin^2 \theta + b^2 \cos^2 \theta$$

TRASFORMAZIONI

DI LORENTZ

$$\begin{cases} ct = x^0 \\ x = x^1 \\ y = x^2 \\ z = x^3 \end{cases}$$

LUNGO
DIREZIONE X

$$\begin{cases} x^0 = \gamma(x^0 - \beta x^1) \\ x^1 = \gamma(x^1 - \beta x^0) \\ x^{2,3} = x^{2,3} \end{cases}$$

LUNGO UNA
DIREZIONE GENERICA

$$\begin{cases} x^0 = \gamma(x^0 - \beta x) \\ x^1 = \gamma(x^1 - \beta x^0) \\ \vec{x}'_{\perp} = \vec{x}_{\perp} \end{cases}$$

ESERCIZIO

(*) A: $x_A = a$ $x_A = z_A = 0$ $t_A = \frac{a}{c}$
B: $x_B = 2a$ $x_B = z_B = 0$ $t_B = \frac{a}{2c}$

DIM. CHE $\exists K'$ TC $t'_A = t'_B$ E TROVA LA
SUA v RISPETTO A K

— 0 —

USARE L'INTERVALLO INVARIANTE

$$(\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

LO CALCOLO

$$(\Delta s_{AB})^2 = -c^2\left(\frac{a}{2c}\right)^2 + (a)^2 = -\frac{a^2}{4} + a^2 = \frac{3}{4}a^2 > 0$$

SE VOLESSI USARE UN ALTRO SR TROVEREI
DEI TERMINI DIVERSI MA COMunque LO
STESSO Δs

Δs^2 HA SEGNO CONCORDO CON Δ IN
TIPO SPAZIO $\Rightarrow$ TIPO SPAZIO

Δs^2 CONCORDO Δt $\Rightarrow$ INTERVALLO TIPO TEMPO

$\Delta s^2 = 0 \Rightarrow$ TIPO LUCE

SE $\Delta t = 0$ IN K'
 $\Rightarrow |\Delta \vec{x}| = l = \sqrt{\Delta s^2} = \frac{\sqrt{3}}{2} a$

ESSENDO $\Delta s^2 > 0$ DUQ $\exists K'$ TC $\Delta t = 0$

CERCA K'

$$x'_A = \gamma(x_A - vt_A) \quad t'_A = \gamma\left(t_A - \frac{v}{c^2}x_A\right)$$

$$x'_B = \gamma(x_B - vt_B) \quad t'_B = \gamma\left(t_B - \frac{v}{c^2}x_B\right)$$

VOGLIO $t'_A = t'_B$

$$\gamma\left(t_A - \frac{v}{c^2}x_A\right) = \gamma\left(t_B - \frac{v}{c^2}x_B\right)$$

$$ct_A - \beta x_A = ct_B - \beta x_B \quad \gamma = \frac{2}{\sqrt{3}}$$

$$a - \beta a = \frac{a}{2} - \beta 2a$$

$$a\beta = \frac{a}{2} \rightarrow \beta = -\frac{1}{2} \rightarrow v = -\frac{c}{2}$$

TROVIAMO

$$t'_A = t'_B = \frac{2}{\sqrt{3}}\left(\frac{a}{c} + \frac{1}{2}\frac{a}{c}\right) = \sqrt{3}\frac{a}{c}$$

$$(C - 5Mx)^2$$

SECONDO $\vec{m} = (\frac{c}{2}, \frac{c}{2}, 0)$

$\vec{v} = (0, \frac{c}{8}, 0)$

ABBINAMO TL (Y)

$$\mu'_x = \frac{m_x}{\gamma(v)(1 - \frac{m_y v}{c^2})}$$

$v = \frac{c}{8}$
 $\gamma = \frac{8}{\sqrt{63}}$
 $\mu'_x = \frac{c/2}{\frac{8}{\sqrt{63}}(1 - \frac{1}{2} \cdot \frac{1}{8})}$
 $= \frac{\sqrt{7}}{5} c$

$$\mu'_y = \frac{m_y - \gamma}{1 - \frac{m_y v}{c^2}}$$

$$= \frac{c/2 - c/8}{(1 - \frac{1}{2} \cdot \frac{1}{8})} = \frac{2}{5} c ; \mu'_z = 0$$

$$\vec{m}' = (\frac{\sqrt{7}}{5} c ; \frac{2}{5} c ; 0)$$

QUADRI-VELOCITÀ $U^M = \gamma(u)(c, \vec{u}) = (\gamma(u)c, \gamma(u)\vec{u})$

K: $|\vec{u}| = \frac{c\sqrt{2}}{2} = \frac{c}{\sqrt{2}}$

$$U^M = \gamma(u)(c, \vec{u}) = \sqrt{2} (c, \frac{c}{2}, \frac{c}{2}, 0) = (\sqrt{2}c, \frac{c}{\sqrt{2}}, \frac{c}{\sqrt{2}}, 0)$$

VALGONO TL (UNGO X)

$$\mu'_x = \frac{m_x - \gamma}{1 - \frac{m_x v}{c^2}}$$

$$\mu'_y = \frac{m_y}{\gamma(v)(1 - \frac{m_x v}{c^2})}$$

ANALOGO A μ'_y CA μ'_z

IN DIREZIONE

$$\mu'_{||} = \frac{m_{||} - \gamma}{1 - \frac{m_{||} v}{c^2}}$$

$$\mu'_{\perp} = \frac{m_{\perp}}{\gamma(v)(1 - \frac{m_{||} v}{c^2})}$$

$$\vec{u} = \frac{\sqrt{2}}{2} c \quad \gamma(u) = \frac{5}{\sqrt{14}}$$

$$U^M = \gamma(u)(c, \vec{u}) = \frac{5}{\sqrt{14}} (\frac{\sqrt{2}}{2} c, \frac{2}{5} c, 0)$$

$$= (\frac{5}{\sqrt{14}} c, \frac{c}{\sqrt{2}}, \frac{\sqrt{2}}{5} c, 0)$$

VERIFICHIAMO CHE U SI TRASFORMA CON TL

$$U'^x = U^x \quad U'^z = U^z$$

$$U'^0 = \gamma(v)(U^0 - \beta U^x) = \frac{8}{\sqrt{63}} (\sqrt{2}c - \beta \frac{c}{\sqrt{2}})$$

$$= \frac{8}{\sqrt{63}} (\sqrt{2}c - \frac{1}{8} \frac{c}{\sqrt{2}}) = \frac{5}{\sqrt{14}} c$$

$$U'^y = \gamma(v)(U^y - \beta U^0) = \frac{8}{\sqrt{63}} (\frac{c}{\sqrt{2}} - \frac{1}{8} \sqrt{2}c) = \frac{\sqrt{2}}{5} c$$

69) $A = (t_A = \frac{2d}{c}; x_A = -d; y_A = 3d; z_A = 0)$

$B = (t_B = \frac{d}{c}; x_B = 2d; y_B = d; z_B = -2d)$

$C = (t_C = 0; x_C = 2d; y_C = d; z_C = 0)$

$d > 0 \quad c = \frac{1}{\sqrt{\epsilon \mu_0}}$

2) IUC EVENTI SONO SIMULTANEI SE

$\Delta s^2 > 0 \quad \text{TIPO SPAZIO}$

$\Delta s^2_{AB} = -(\bar{x}_B - \bar{x}_A)^2 + (\bar{x}_3 - \bar{x}_A)^2$

3) CALCOLO E Δs^2 VERO QUALE EVENTI, DANNO UN TIPO SPAZIO

$\Delta s^2_{AB} = -(-d - 2d)^2 + ((2d - 2d)^2 + (d - 3d)^2 + (-2d)^2)$
 $= -9d^2 + 8d^2 = -d^2 < 0$

$\Delta s^2_{BC} = -(-d - 0)^2 + ((2d - 2d)^2 + (d - d)^2 + (-2d)^2)$
 $= -d^2 + 4d^2 = 3d^2 > 0$

$\Delta s^2_{AC} = -(2d)^2 + ((2d - 2d)^2 + (3d - d)^2 + 0)$
 $= -4d^2 + 4d^2 = 0$

QUINDI:	Δs_{AB}	TIPO	TEMPO
	Δs_{BC}	TIPO	SPAZIO
	Δs_{AC}	TIPO	LUCE

4) $t_B = t_C \Rightarrow \Delta t_{BC} = 0 \quad \text{avendo}$

$\Delta s^2 = -(\Delta x')^2 = 3d^2 \Rightarrow \Delta x' = \sqrt{3}d$

b) AB POSSONO ESSERE IN UN K' $\text{t.c. } |\Delta x'| = 0$
 ESSENDO $\Delta s^2 < 0$

SE $|\Delta x'| = 0 \Rightarrow \Delta s^2 = -(\Delta t')^2 = -d^2$
 $\Rightarrow c \Delta t' = d \Rightarrow \Delta t' = d/c$

c) LA TERZA COPPIA DI EVENTI AC CON $\Delta s_{AC} = 0$ È UN INTERVALLO DI TIPO LUCE E POSSONO INFLUENZARSI SOLO CON L'INVIO DI SEGNALE LUMINOSI
 E A K' $\text{t.c. } t'_A = t'_C \quad \Delta x' = 0$

d) SOLO AB E AC

e) NO $t_B < t_C$ IN K

MA $t_B < t_A$ IN K'

$\Delta s^2_{BC} = 3d^2$

IN K' $\begin{cases} t'_B = \gamma(t_B - \frac{v}{c} z_B) \\ t'_C = \gamma(t_C - \frac{v}{c} z_C) \end{cases}$

$\Delta t'_{BC} > 0 \quad \text{poiché } \Delta t_{BC} = -d < 0$

$\Delta t'_{AC} = \gamma(t_A - t_C + \frac{v}{c} 2d) > 0$

$\Delta t'_{AC} = \beta 2d > 0 \Rightarrow \beta > -\frac{t_C}{2d} = \frac{1}{2}$

E SO $\beta < 1 \Rightarrow \boxed{\frac{1}{2} < \beta < 1}$

$(c + 5Mx)^2$

66) A CAUSA DI B ; B CAUSA DI C

$$\begin{cases} A: t_A = \frac{B}{C} & x_A = 2 & y_A = z_A = 0 \\ B: t_B = \frac{A}{C} & x_B = 2B & y_B = z_B = 0 \\ C: t_C = \frac{A^2}{C} & x_C = \frac{A}{C} & y_C = z_C = 0 \end{cases}$$

AVO A SO K' TC $\vec{r} = \frac{B}{C} C$ LUNGO X
E $x'_A = x'_B$

MA E SO BC TIPO LUCE
O

CALCOLO $\Delta s_{BC}^2 = -c^2(t_B - \frac{4B}{C})^2 + (2B - x_C)^2 = 0$
 $= 0 - c^2(t_B - \frac{4B}{C})^2 + (2B - x_C)^2 = 0$

CALCOLO $\Delta s_{AB}^2 = -c^2(\frac{B}{C} - t_A)^2 + (2 - 2A)^2$
 $= -c^2(\frac{B}{C} - t_A)^2 + 2^2$
SE $\exists K' TC$ $x'_A = x'_B$ ALLORA $\Delta s_{AB}^2 < 0$
 $-c^2(\frac{B}{C} - t_A)^2 + 2^2 < 0$
 $-c^2(\frac{B^2}{C^2} - 2\frac{B}{C}t_A + t_A^2) + 2^2 < 0$
 $-2^2 + 2ac t_B - c^2 t_B^2 + 2^2 < 0$

OPPURE $x'_A = \gamma(x_A - \beta c t_A)$
 $x'_B = \gamma(x_B - \beta c t_B)$

$\gamma(x_A - \beta c t_A) = \gamma(x_B - \beta c t_B) \Rightarrow t_B = \frac{\alpha\beta + x_B - x_A}{\beta c} =$

$$b_0 = \frac{2\frac{2}{3} + 2a - a}{\frac{2}{3}c} = \frac{5a}{3c} \left(\frac{2}{5} + \frac{10}{5} - \frac{5}{5} \right) = \frac{4a}{3c}$$

SOSTITUIENDO $\Delta s_{bc}^2 = 0$

$$-c^2 \left(\frac{4a}{3c} - 4\frac{a}{c} \right)^2 + 4a^2 - 4ax_c + x_c^2 = 0$$

$$-\frac{16}{9}a^2 + 4a^2 - 4ax_c + x_c^2 = 0$$

$$x_c^2 - 4ax_c + \frac{20}{9}a^2 = 0$$

$$x_c = \frac{4a \pm \sqrt{16a^2 - 4\frac{20}{9}a^2}}{2} = 2a \pm \sqrt{4a^2 - \frac{20}{9}a^2}$$

$$= 2a \pm \sqrt{\frac{36-20}{9}}a = 2a \pm a\sqrt{\frac{16}{9}} = \left(2 \pm \frac{4}{3}\right)a$$

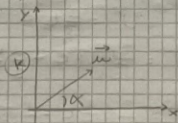
$$x_c = \begin{cases} \frac{10}{3}a \\ +\frac{2}{3}a \end{cases}$$

5° IN K SI MUOVE SU XY CON $\vec{u}$

K' $\vec{r} = r\hat{x}$ RISPETTO K

IN $\vec{u} = u\hat{x}$

$$\begin{cases} u = 0,80 \\ v = 0,40 \end{cases}$$



vale $\begin{cases} u_x = u \cos \alpha \\ u_y = u \sin \alpha \end{cases}$

USO LA COMPOSIZIONE DELLE VELOCITA'

$$u'_x = \frac{u_x - v}{1 - v \frac{u_x}{c^2}} = 0$$

$$u'_y = \frac{u_y}{\gamma(1 - v \frac{u_x}{c^2})}$$

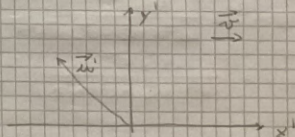
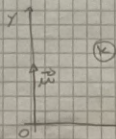
$$u'_x = 0 = \frac{u \cos \alpha - v}{1 - v \frac{u \cos \alpha}{c^2}} = 0 \Rightarrow \alpha = \cos^{-1} \left(\frac{v}{u} \right) = \frac{\pi}{3}$$

quindi

$$u'_y = \frac{u_y}{\gamma(1 - v \frac{u_x}{c^2})} = 2,36 \cdot 10^8 \frac{m}{s} = 0,9c$$

68 $\vec{u} = (0, \frac{c}{\sqrt{3}}, 0)$ IN K

IN K' HA $\vec{u} = u\hat{x}$ P $\vec{u}$ HA $\alpha = 35^\circ$



$$(c = 3u_x)$$

K' AVRE

$$\begin{cases} \dot{m}_x' = \dot{m} \cos \alpha \\ \dot{m}_y' = \dot{m} \sin \alpha \end{cases}$$

$$\alpha = \frac{3\pi}{4}$$

E IN K HO

$$\begin{cases} \dot{m}_x = 0 \\ \dot{m}_y = c/\sqrt{3} \end{cases}$$

USO LA COMPOSIZIONE DELLE VELOCITA

$$\dot{m}_x = \frac{\dot{m}_x - v}{1 - v \frac{\dot{m}_x}{c^2}} = -v$$

$$\dot{m}_y = \frac{\dot{m}_y}{\gamma(1 - v \frac{\dot{m}_x}{c^2})} = \frac{\dot{m}_y}{\gamma} \rightarrow \gamma = \frac{\dot{m}_y}{\dot{m}_y'}$$

$$\rightarrow \sqrt{1 - \beta^2} = \frac{\dot{m}_y}{\dot{m}_y'} \rightarrow \beta^2 = 1 - \left(\frac{\dot{m}_y}{\dot{m}_y'}\right)^2$$

$$v = c \sqrt{1 - \left(\frac{\dot{m}_y}{\dot{m}_y'}\right)^2}$$

$$-\dot{m}_x = c \sqrt{1 - \left(\frac{\dot{m}_y}{\dot{m}_y'}\right)^2} \rightarrow (\dot{m}_x')^2 = c^2 \left(1 - \left(\frac{\dot{m}_y}{\dot{m}_y'}\right)^2\right)$$

$$(\dot{m}_x')^2 \left(\cos^2 \alpha + c^2 \frac{\dot{m}_x'^2}{\dot{m}_y'^2} \right) = c^2$$

$$(\dot{m}_x')^2 = \frac{\dot{m}_y'^2 c^2}{\cos^2 \alpha + c^2 \frac{\dot{m}_x'^2}{\dot{m}_y'^2}}$$

$$\dot{m} = \left(\frac{\dot{m}_y'^2 c^2}{\cos^2 \alpha + c^2 \frac{\dot{m}_x'^2}{\dot{m}_y'^2}} \right)^{1/2} = 7,07 \cdot 10^7 = 0,235 c \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{6}}{6} c$$

ALLORA TROVO

$$\dot{m}_y' = \dot{m} \sin \alpha = \frac{\sqrt{2}}{6} c \frac{\sqrt{3}}{2} = c/6$$

$$\dot{m}_x' = \dot{m} \cos \alpha = -c/6$$

$$\dot{m}_x = -v \Rightarrow v = c/6$$

69) IN (K) $\vec{u}_A = (\frac{2}{3}c, \frac{1}{3}c, \frac{1}{3}c)$

(K) CON $\vec{v} = (0, \frac{c}{2}, 0)$

$$|\vec{u}_A| = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{1}{9}} c = \frac{\sqrt{6}}{3} c$$

$$\gamma(u) = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \sqrt{3} \quad \gamma(v) = \frac{1}{\sqrt{1 - \frac{1}{4}}} = \sqrt{5}$$

$$U^M = \sqrt{5} \left(c, \frac{2}{3}c, \frac{c}{3}, \frac{c}{3} \right) = \left(\sqrt{5}c, \frac{2\sqrt{5}}{3}c, \frac{\sqrt{5}}{3}c, \frac{\sqrt{5}}{3}c \right)$$

HO UN COORDINATO

$$U^1 = U^1 = \frac{2}{\sqrt{3}} c \quad U^4 = U^4 = \frac{c}{\sqrt{3}}$$

$$U^0 = \frac{2}{\sqrt{3}} \left(\sqrt{5}c - \frac{1}{2} \frac{\sqrt{5}}{3} c \right) = 2c \left(1 - \frac{1}{6} \right) = \frac{5}{3} c$$

$$U^2 = \frac{2}{\sqrt{3}} \left(\frac{1}{3}c - \frac{1}{2} \frac{\sqrt{5}}{3} c \right) = c \left(\frac{2}{3} - 1 \right) = -\frac{c}{3}$$

QUINDI $U^M = \left(\frac{5}{3}c, \frac{2}{3}c, -\frac{c}{3}, \frac{c}{\sqrt{3}} \right)$

30 $(1\beta^2 = -c^2 - (-1)^2 + 1^2 + 1^2)$

$B^\mu = (0, 0, 1, 0)$ TIPO SPAZIO
CHE SCEGLIAMO

$A^\mu B_\mu = 0$ SODDISFATTO
PER ESEMPPIO $A^\mu = (0, 1, 0, 0)$ CHE
NON TIPO TEMPO

(*)
$$\frac{c^2(1-\beta^2) - m_x^2(1-\beta^2) - (m_y^2 + m_z^2)(1-\beta^2)}{(c - \beta m_x)^2}$$

$$\frac{(1-\beta^2)(c^2 - m_x^2 - m_y^2 - m_z^2)}{(c - \beta m_x)^2}$$

$$\frac{(1-\beta^2)(c^2 - m^2)}{(c - \beta m_x)^2} > 0 \Leftrightarrow c^2 - m^2 > 0$$

$$\Leftrightarrow c^2 > m^2$$

$$\Leftrightarrow c > m$$

PRENDO $\mu = (m_x, m_y, m_z)$ $v = (v, 0, 0)$

FACCIO $1 - \frac{m^2}{c^2} > 0$

BOOST LUNGO X

$$\begin{cases} m'_x = \frac{m_x - v}{1 - \frac{vm_x}{c^2}} \\ m'_y = \frac{m_y}{\gamma(1 - \frac{vm_x}{c^2})} \\ m'_z = \frac{m_z}{\gamma(1 - \frac{vm_x}{c^2})} \end{cases}$$

CALCOLO

$$(m')^2 = (m'_x)^2 + (m'_y)^2 + (m'_z)^2$$

$$= \frac{(m_x - v)^2 + (m_y^2 + m_z^2)(1 - \beta^2)}{(1 - \frac{vm_x}{c^2})^2}$$

$$1 - \frac{m'^2}{c^2} = 1 - \frac{(m_x - v)^2 + (m_y^2 + m_z^2)(1 - \beta^2)}{c^2(1 - \frac{vm_x}{c^2})^2}$$

$$c^2 \left(1 - \frac{vm_x}{c^2}\right)^2 = (m_x - v)^2 + (m_y^2 + m_z^2)(1 - \beta^2)$$

$$\frac{c^2 \left(1 - \frac{vm_x}{c^2}\right)^2}{(c - \beta m_x)^2} = \frac{(m_x - \beta c)^2 + (m_y^2 + m_z^2)(1 - \beta^2)}{(c - \beta m_x)^2}$$

(*)
$$\frac{c^2 + \beta^2 m_x^2 - 2\beta c m_x - m^2 - \beta^2 c^2 + 2\beta c m_x + (m_y^2 + m_z^2)(1 - \beta^2)}{(c - \beta m_x)^2}$$

$$\odot \quad \text{so} \quad (U^0)^2 = -c^2 \quad \text{--} \quad U^0 + U^1 + U^2 + U^3$$

$$-4c^2 + \frac{3}{4}c^2 + U^3^2 = -c^2$$

$$U^3^2 = c^2 \left(-1 + 4 - \frac{3}{4} \right) = \frac{9}{4}c^2$$

$$\Rightarrow U^3 = \frac{3}{2}c$$

allora $U^3 = \frac{3}{4}c$

$$\Rightarrow U^\mu = (2c, \frac{\sqrt{3}}{2}c, 0, \frac{3}{2}c)$$

$$U^\mu = (\frac{\sqrt{3}}{4}, 0, \frac{3}{4}c)$$

b) so $U^3 = v$ dalla composizione delle velocità

$$\Rightarrow v = \frac{3}{4}c$$

allora $\beta = \frac{3}{4} \Rightarrow \mu = \tanh^{-1} \beta = 0,972$

c) ho usato Lorentz su U^μ e ho trovato

$$U'^\mu = (\gamma(v)(2c - \frac{\sqrt{3}}{2}U^3), \frac{\sqrt{3}}{2}c, 0, 0)$$

$$\frac{1}{1 - \frac{v^2}{c^2}} \left(2c - \frac{3}{4}c \right) = \frac{4}{\sqrt{7}} \frac{7c}{8} = \frac{7c}{2\sqrt{7}} = \frac{\sqrt{7}}{2}c$$

$$U'^\mu = \left(\frac{\sqrt{7}}{2}c, \frac{\sqrt{3}}{2}c, 0, 0 \right)$$

$$\boxed{7.1} \quad \odot \quad U^\mu = (U^0, \vec{U}) = (2c, \frac{\sqrt{3}}{2}c, 0, U^3)$$

$$\odot \quad \text{ha} \quad \vec{U} = v \hat{z} \quad \text{e} \quad P \quad \text{si} \quad \text{muove} \quad \text{lungo}$$

$$\odot \quad U^\mu = (\gamma(v)c, \gamma(v)\vec{U})$$

allora $U^0 = \gamma(v)c \Rightarrow \gamma(v) = 2$

$$U^1 = \gamma(v)U^1 \Rightarrow U^1 = \frac{\sqrt{3}}{4}c$$

$$U^2 = \gamma(v)U^2 \Rightarrow U^2 = 0$$

$$U^3 = \gamma(v)U^3 = 2U^3 \Rightarrow U^3 = \frac{U^3}{2}$$

compongo le velocità

$$U^3 = \frac{U^3 - v}{1 - \frac{vU^3}{c^2}} = 0 \Rightarrow U^3 = v$$

$$U^1 = \frac{U^1}{\gamma(1 - \frac{vU^3}{c^2})} = \frac{U^1}{\gamma(1 - \frac{v^2}{c^2})} = \gamma(v)U^1$$

$$U^2 = 0$$

uso Lorentz

$$U^1 = U^1 = \frac{\sqrt{3}}{2}c \quad U^1 = U^2 = 0$$

$$U^3 = \gamma(v) \left(U^3 - \frac{v}{c} U^0 \right) = \gamma(v) \left(U^3 - 2v \right)$$

$$= \gamma(v) (2U^3 - 2v) = \gamma(v) (2U^3 - 2U^3) = 0$$

$$U^0 = \gamma(v) \left(U^0 - \frac{v}{c} U^3 \right)$$

ARGENTINA

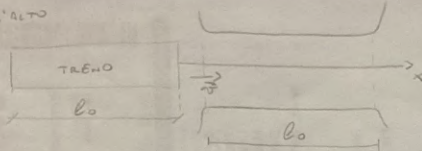
South Georgia

Gough Island

French Southern and Antarctic Lands (FRANCE)

Prince Edward Islands SOUTH AFRICA

#2 CALLATO



(*) SR GALLERIA $\rightarrow$ TRENO CONTRATTO

(*) SR TRENO $\rightarrow$ GALLERIA CONTRATTA

$$x = x' = 0 \quad t = t' = 0$$

— O —

$$\begin{array}{l|l} K & \text{PUNTA} \\ x_p = vt & \\ x_c = vt - l & \text{CODA} \end{array}$$

$$\text{so } l = \frac{l_0}{\gamma} \rightarrow x_c = vt - \frac{l_0}{\gamma}$$

QUANDO LA CODA ENTRA $x_c = 0 \quad t = t_1$

$$\Rightarrow 0 = vt_1 - \frac{l_0}{\gamma} \quad \text{e} \quad x_p = l = vt_1$$

$$\Rightarrow vt_1 = \frac{l_0}{\gamma} \Rightarrow t_1 = \frac{l_0}{\gamma v}$$

QUINDI A t_1 LA CODA SI TROVA
A $x=0$ E LA PUNTA A DISTANZA
 $l = \frac{l_0}{\gamma}$

QUANDO LA TESTA ESCE $x_p = l_0 \quad t = t_2$
 $\Rightarrow l_0 = vt_2 \Rightarrow t_2 = \frac{l_0}{v}$

E NELLO STESSO TEMPO LA CODA
SI TROVA A $x_c = vt_2 - l = l_0 - l$

$$\begin{array}{l|l} K' & \text{OUT} \\ x'_0 = -vt' + l = -vt' + \frac{l_0}{\gamma} & \\ x'_1 = -vt' & \text{IN} \end{array}$$

TESTA ESCE $\Rightarrow x'_0 = 0 \quad \text{e} \quad t'_1 = t'_2$

$$0 = -vt'_2 + l \rightarrow t'_2 = \frac{l_0}{\gamma v}$$

$$x'_2(t'_2) = -vt'_2 = -\frac{l_0}{\gamma}$$

LA CODA ENTRA $\Rightarrow x'_1 = -l_0 \quad \text{e} \quad t'_1 = t'_1$

$$-l_0 = -vt'_1 \Rightarrow t'_1 = \frac{l_0}{v}$$

$$x'_0(t'_1) = v \frac{l_0}{v} + \frac{l_0}{\gamma} = l_0 \left(1 + \frac{1}{\gamma}\right)$$

CON LORENZ

$$t'_1 = \gamma \left(t_1 - \frac{v}{c^2} x_1 \right) = \gamma \left(\frac{l_0}{\gamma v} - \frac{v}{c^2} 0 \right) = \frac{l_0}{v}$$

$$\begin{aligned} t'_2 &= \gamma \left(t_2 - \frac{v}{c^2} x_2 \right) = \gamma \left(\frac{l_0}{v} - \frac{v}{c^2} l_0 \right) \\ &= \frac{l_0}{\gamma v} \left(\gamma^2 \left(1 - \frac{v^2}{c^2} \right) \right) = \frac{l_0}{\gamma v} \left(\gamma^2 \frac{1}{\gamma^2} \right) = \frac{l_0}{\gamma v} \end{aligned}$$

NOTA $t_2 > t_1 \quad \text{e} \quad t'_1 > t'_2$

QUINDI IN K LA TESTA ESCE DOPO CHE
LA CODA È ENTRATA QUINDI C'È UN
MOMENTO IN CUI TUTTO IL TRENO
È IN GALLERIA.

IN K' LA TESTA ESCE PRIMA CHE LA
CODA SIA ENTRATA.

PER I TEMPI K E K' VEDONO COSÌ
DIVERSE MA IL RISULTATO È SEMPRE
CHE IL TRENO ATTRAVERSA LA GALLERIA

DIAGRAMMI DI MINNOWSKI

